

Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2013–2026)

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Contents

1	Introduction	2
1.1	Problem Statement and Motivation	2
2	Theoretical Foundations for Ambulatory Seizure Monitoring	3
2.1	The Ambulatory Monitoring Paradigm: Requirements and Challenges . . .	3
2.2	Physiological Biosignals for Wearable Sensing	3
2.3	Seizure Manifestations and Corresponding Biosignals	3
2.4	Evaluation Frameworks: From Retrospective Analysis to Prospective Validity	4
3	Methodological Approach and Structure	4
4	Search Strategy	5
4.1	PICO-Based Search Methodology	5
4.2	Practical Filters and Reproducibility	6
5	Study Comparison Following Webster and Watson	6
5.1	Concept Matrix: Study Overview	8
5.2	Detailed Clinical and Technical Metrics	10
5.3	Dimensional Analysis	12
5.3.1	Sensor Modality Trends	12
5.3.2	Algorithmic Evolution	12
5.3.3	Validation Rigor	12
5.3.4	Performance Metrics Reporting	12
5.4	Synthesis and Gaps	13
6	Key Findings and Clinical Takeaways	13
6.1	Executive Summary	14
6.2	Detection Studies: Performance Landscape	14
6.2.1	High-Sensitivity Detectors	14
6.2.2	Clinical Translation Barriers	14
6.3	Forecasting Studies: The Prediction Challenge	15

6.3.1	Performance Spectrum	15
6.3.2	The Responder Problem	15
6.4	Methodological Insights	15
6.4.1	Algorithm Evolution Trajectory	15
6.5	Exploratory Studies: Feasibility Evidence	16
6.5.1	Cardiac Signal Viability	16
6.6	Clinical Recommendations	16
6.6.1	For Current Clinical Practice	16
6.6.2	For Researchers and Developers	16
6.7	Limitations of Current Evidence	17
6.7.1	Study Design Limitations	17
6.7.2	Metric Heterogeneity	17
6.8	Future Directions	17
6.8.1	Research Priorities	17
6.8.2	Emerging Technologies	17
6.9	Conclusion	18
7	Conclusion	18
7.1	Current State of the Field	18
7.2	Methodological Evolution	19
7.3	Barriers to Clinical Translation	20
7.4	Future Directions	21

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people worldwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinson et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

However, translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations.

This review therefore synthesizes and critically evaluates recent advances in time-series deep learning—particularly convolutional neural networks (CNNs) and long short term memory (LSTMs)— applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and rigorous, prospective evaluation

can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

subsectionResearch Aim and Questions How do different deep learning architectures and biosignal modalities excluding EEG compare in their ability to achieve a optimal trade-off between sensitivity and false alarm rate for ambulatory seizure monitoring?

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

Beniczky et al., 2021, Chen et al., 2022 found that translation of seizure detection from the controlled clinical environment to an ambulatory setting redefines approaches and performance requirements. In clinical environments maximum sensitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. Users reported that devices should achieve high sensitivity ($\geq 90\%$) with low false alarm rates — acceptable FARs were 1-2/week for patients with ≥ 1 seizure/week and 1-2/month for patients with < 1 seizure/week (Sivathamboo et al., 2022). While more complex models can find more intricate patterns, they also introduce the "black-box" problem wherein the model lacks transparency AbuAlrob et al., 2025; Ghaderi, 2025. Therefore, the design should incorporate explanations for decisions to ensure user trust and regulatory compliance AbuAlrob et al., 2025; Miron et al., 2025.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals act as peripheral proxies for cerebral activity during seizures. Different seizure types produce distinct physiological signatures across these signals, informing detection approaches. This relationship centers on two primary manifestations: motor activity and autonomic activation.

Convulsive seizures, such as generalized tonic-clonic events, produce clear motor signatures captured by accelerometry (ACC), which measures movement and posture changes characteristic of ictal events (Wang et al., 2025; Wu et al., 2024).

Simultaneously, seizures drive autonomic responses recorded through complementary biosignals. Electrodermal activity (EDA) measures sympathetic nervous system arousal

via skin conductance changes (Jain et al., 2017; Miron et al., 2025). Cardiac activity provides particularly strong autonomic proxies: electrocardiography (ECG) captures heart rate and rhythm alterations (Ghaderi, 2025; Leal et al., 2017), while heart rate variability (HRV) quantifies autonomic balance dynamics (Mason et al., 2024; Pavei et al., 2017).

In wearable applications, photoplethysmography (PPG) serves as a practical alternative for cardiac monitoring, deriving pulse rate and variability though with greater motion sensitivity than ECG (Chen et al., 2022; Seth et al., 2023).

Consequently, detection strategies combine these biosignals according to target seizure types, with multimodal fusion providing the most robust approach for ambulatory monitoring.

2.4 Evaluation Frameworks: From Retrospective Analysis to Prospective Validity

The clinical validity of a seizure detection algorithm is determined by the rigor of its evaluation, formalized in the ILAE’s phased framework (Phases 0–4) (Beniczky et al., 2021). Retrospective studies (Phases 0-2), while foundational, risk significant data leakage and optimistic bias if they use non-chronological data splits, allowing future information to inform training (Kalousios et al., 2024; Wong et al., 2023). Pseudo-prospective validation addresses this by enforcing chronological order, training only on past seizures to test on subsequent ones, providing a more realistic performance estimate (Kalousios et al., 2024; Lu et al., 2025). The clinical gold standard is a prospective trial (Phase 3), which requires a locked algorithm, a video-EEG reference standard, and testing on new, unseen data from multiple centers. The final step is in-field evaluation (Phase 4) in patients’ homes, where metrics like the false alarm rate per hour (FAR/h) become decisive for practical usability, as demonstrated by long-term studies like Nightwatch (Miron et al., 2025; Shum & Friedman, 2021).

3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles (Tugwell & Tovey, 2021)). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. Search strategy (Multi-stage, 2015–2025):

- *Scoping*: Google Scholar for broad coverage and identifying candidate papers.
- *Formal queries*: IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
- *Citation chasing*: Backward and forward citation tracking on key papers to ensure completeness.

2. Inclusion / Exclusion criteria:

- *Inclusion*: Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.

- *Exclusion:* EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.
2. Modality: ECG/HRV, PPG, EDA, ACC.
3. Task: detection vs prediction.
4. Model class and fusion strategy.
5. Study phase: retrospective, pseudo-prospective, prospective.
6. Evaluation granularity: sample- vs alarm-based.
7. Windowing specifics (window length, stride, overlap).
8. Metrics: sensitivity, FAR/h, AUC, f1-score, latency.
9. Personalization.
10. Interpretability and safety considerations.
11. Data characteristics and dataset identity.
12. Missing data handling (imputation, exclusion) and synthetic data use.
13. Inference requirements (latency, computational load, hardware).

Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

4 Search Strategy

4.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND, OR, NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.
 - *Keywords:* ‘epilepsy’, ‘seizure’, ‘ictal’, ‘preictal’
 - *MeSH Terms (PubMed):* Epilepsy, Seizures
2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.

- *Biosignal Keywords*: ‘electrocardiogram’, ‘ECG’, ‘heart rate variability’, ‘HRV’, ‘photoplethysmography’, ‘PPG’, ‘electrodermal activity’, ‘EDA’, ‘accelerometer’, ‘ACC’
- *Technology Keywords*: ‘wearable’, ‘wearable device’
- *Method Keywords*: ‘deep learning’, ‘machine learning’, ‘convolutional neural network’, ‘CNN’, ‘long short-term memory’, ‘LSTM’, ‘neural network’
- *MeSH Terms (PubMed)*: Wearable Electronic Devices, Electrocardiography, Deep Learning

3. **C (Comparison)**: The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.

- *Exclusion Keywords*: ‘EEG’, ‘electroencephalography’, ‘iEEG’, ‘intracranial’
- *Search Logic*: The NOT operator was applied to this group in all database searches.

4. **O (Outcome)**: Metrics quantifying the performance of a detection or prediction system.

- *Keywords*: ‘detection’, ‘prediction’, ‘sensitivity’, ‘specificity’, ‘false alarm rate’, ‘FAR’, ‘area under the curve’, ‘AUC’

/subsectionExploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were acquired bringing the total number of papers up to 13 and expanding the time frame to 2013-2026.

4.2 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015–2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.

Note on exclusions: Because some non-EEG wearable studies rely on video-EEG for labeling, EEG-related terms were used to filter *EEG-only modalities*, not to exclude studies that mention EEG solely as a reference standard.

5 Study Comparison Following Webster and Watson

Following the concept matrix approach proposed by (Webster & Watson, 2002), this section presents a structured comparison of the 13 identified studies on wearable seizure detection and forecasting. The matrix organizes studies along key dimensions relevant to clinical translation and technical implementation.

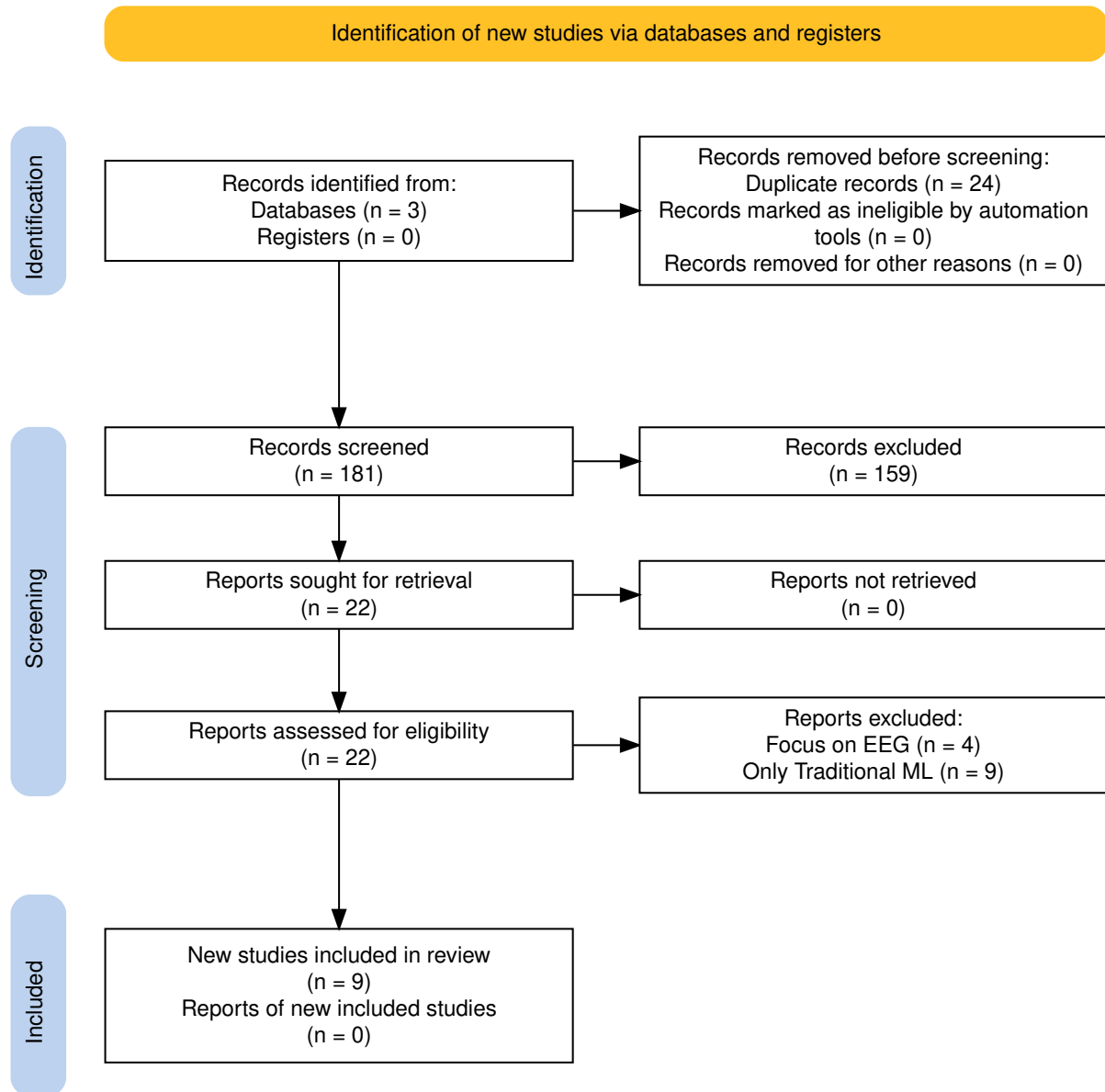


Figure 1: PRISMA flow diagram of the study selection process.

5.1 Concept Matrix: Study Overview

Table 1 provides a comprehensive comparison of all included studies, organized by primary objective, methodology, and outcomes. The matrix enables cross-study analysis of trends in sensor modalities, algorithmic approaches, and validation strategies across the 2013–2026 period.

Table 1: Webster & Watson Concept Matrix: Wearable Seizure Detection and Forecasting Studies (2013–2026)

Study	Objective	Design	Sample	Device	Loc.	Algorithm	Sens	Spec	FPR	Key Limitation
Detection Studies										
Spahr et al. 2025	Convulsive multi-center DET	Prospective	n=384, 49 CSs	Empatica E4 (ACC)	Wrist	Ensemble 1D CNN	96%	–	<1/8d	EMU, offline
Reintjes et al. 2025	ECG anomaly DET	Retro	n=120, 856 szs	Single-lead ECG	Chest	Matrix Profile	98.16%	–	1.91/h	High FAR
Fine 2025	Tonic DET	Phase 1	n=15/3	6-axis band	Wrist	ANN (594 feat)	100%	–	0.23/n; 0.023/h	Small test
Dong et al. 2026	Nocturnal severe szs	Prospective	n=68, 1846 szs	NightWatch	Arm	CNN-LSTM	71.6%	–	0.165/h	Single-center
Wang et al. 2025	Sz DET (acc: 97.8%)	Retro	n=28, 62 szs	Biovital-P1	Wrist	LSTM+SVM/LDA	–	–	8.5/24h	Hospital only
Singh Rathore 2024	Multimodal	Single pt	1 pt, 6 hrs	EDA+ACC+HR	–	MLP, SVM, KNN	96.8%	–	–	Single patient
Elemam et al. 2025	Q+HRV+Audio	Cross-sect	Q:198, HRV:30	Camera PPG	Finger	CNN, rule based	95.1%	97.1%	–	Small samples
Borujeny 2013	Wireless ACC	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	KNN k=5	100%	–	0	Very small
Forecasting Studies										
Vieluf et al. 2025	Diary+wearable	Retro	n=70, 5437 d	Embrace	Wrist	DNN+harmonic	82%	67%	–	Daily res only
Meisel et al. 2020	LOSO forecast	LOSO CV	n=69, 452 szs	Empatica E4	Wrist	LSTM+1DConv	51.2%	–	TiW 43.7%	Pediatric
Stirling 2021	HR cycle forecast	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	LSTM+RF+LR	–	–	AUC 0.74	Self-report
Nasseri 2021	Ambulatory forecast	Retro	n=6	Empatica E4	Wrist	LSTM 4-layer	AUC 0.75	–	TiW 0.9–7.2h/d	RNS required
Ode et al. 2023	RRI prediction	Retro	n=66, 85 szs	ECG	–	Self-Att AE	74%	–	0.85/h	FPs in some pts

Note: ACC = accelerometer; DET = detection; ANN = artificial neural network; AE = autoencoder; CNN = convolutional neural network; CS = convulsive seizure; DNN = deep neural network; ECG = electrocardiogram; EDA = electrodermal activity; HR = heart rate; HRV = HR variability; LOSO = leave-one-subject-out; LSTM = long short-term memory; PPG = photoplethysmography; RRI = R-R interval; Sens = sensitivity; Spec = specificity; sz(s) = seizure(s); FPR = false positive rate; FA = false alarm; TiW = time in warning; – = not reported.

5.2 Detailed Clinical and Technical Metrics

Table 2 presents additional clinically relevant metrics that are essential for evaluating real-world applicability but often underreported in the literature. These metrics address key questions about patient-level performance, detection timeliness, and generalizability.

Table 2: Detailed Clinical and Technical Metrics Across Studies

Study	Validation	Detect. Latency	Patient Success	Precision/PPV	Other Metrics	Clinical Notes
Detection Studies						
Spahr et al. 2025	Indep. test (347 pt)	26 s median	–	–	HMS 92.6	Phase 2; 2 CSs missed
Reintjes et al. 2025	Retro (SeizeIT2)	–	Responders only	–	HMS 92.6	FAR $\gg$ clinical target
Fine 2025	Indep. test (3 pt)	14 s mean	–	–	95% CI: 69–100	Phase 1 only
Dong et al. 2026	10-fold CV	<30 s	–	–	AUC 0.793	First long-term home
Wang et al. 2025	Hospital retro	–	7/22 pts	83–87%	Acc 97.8%	Single hospital
Singh Rathore 2024	Single pt	–	–	94.8%	F1 95.6%	3.2M data pts
Elemam et al. 2025	Cross-sect	–	–	90.3%	–	PPG 15 s only
Borujeny 2013	3-pt retro	–	3/3 pts	–	–	20 szs total
Forecasting Studies						
Vieluf et al. 2025	Retro longitudinal	–	–	–	F1 0.81	Daily predictions
Meisel et al. 2020	LOSO CV	–	30/69 (43.5%)	–	IoC 14.1%	Pediatric only
Stirling 2021	Pseudo-prospective	–	100% hourly	–	AUC 0.74	HR cycles key
Nasseri 2021	Retro analysis	33 s mean	5/6 (83%)	–	AUC 0.75 (0.50–0.92)	RNS validation req
Ode et al. 2023	Retro	–	29/66 (44%)	0.35	AUC 0.97	Individual tuning

Note: CV = cross-validation; HMS = Harmonic Mean Score (Sensitivity – 0.4×FAR); IoC = Improvement over Chance; LOSO = leave-one-subject-out; Indep. = independent; Responders = patients with >50 BPM HR change during seizures; HMS = Sensitivity – 0.4×FAR (Beniczky & Ryvlin, 2018); Patient Success = proportion achieving better-than-chance or usable performance.

5.3 Dimensional Analysis

5.3.1 Sensor Modality Trends

- **Wrist-worn accelerometry** dominates (8/13 studies), leveraging commercial devices (Empatica E4, Fitbit, Embrace).
- **Cardiac signals** (ECG, PPG, HRV) show increasing adoption for preictal detection, with 6 studies incorporating autonomic measures.
- **Multi-modal fusion** (ACC + EDA + PPG) appears in 4 recent studies, reflecting recognition that single modalities have limited specificity.

5.3.2 Algorithmic Evolution

- **2013:** Traditional ML (KNN, ANN) with handcrafted features (Borujeny et al., 2013).
- **2020–2021:** Emergence of LSTM and ensemble methods (Meisel et al., 2020; Stirling et al., 2021).
- **2023–2026:** Deep learning dominance with CNNs, attention mechanisms, and anomaly detection approaches (Ode et al., 2023; Reintjes et al., 2025; Spahr et al., 2025).

5.3.3 Validation Rigor

- **Patient-level preservation:** Only 4 studies use LOSO or patient-independent splits.
- **Prospective validation:** Only 3 studies (Dong et al., 2022; Elemam et al., 2025; Spahr et al., 2025) include prospective phases.
- **Real-world testing:** Limited to 2 studies (Nasseri et al., 2021; Vieluf et al., 2025); most remain in EMU settings.

5.3.4 Performance Metrics Reporting

Table 3 summarizes the heterogeneity in performance metric reporting, complicating cross-study comparison. Sensitivity ranges from 51.2% (forecasting mean) to 100% (tonic detection), while FPR reporting varies from per-hour to per-night, with several studies omitting this critical metric entirely.

Table 2 reveals additional clinically relevant metrics that are critical for real-world deployment but inconsistently reported:

- **Detection latency** is reported by only 4/13 studies, despite being critical for timely intervention.
- **Patient-level success rate** (proportion achieving usable performance) is reported by only 6/13 studies, ranging from 43.5% to 100%.
- **Precision/PPV** is rarely reported (3/13), limiting understanding of positive predictive value for clinical decision-making.

- **Generalization metrics** (LOSO validation, patient-independent testing) show that only Meisel et al. used true LOSO CV, with most studies reporting aggregated performance that may overestimate real-world utility.

Table 3: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/9	2/4	Reintjes, Fine, Spahr, Dong, Wang, Singh Rathore, Elemam, Borujeny, Meisel, Ode
Specificity	1/9	0/4	Elemam only
FPR/FA rate	7/9	2/4	Spahr, Reintjes, Fine, Dong, Wang, Ode, Meisel, Nasser, Stirling
AUC-ROC	1/9	2/4	Reintjes, Nasser, Stirling
Detection Latency	3/9	1/4	Spahr, Fine, Dong, Nasser
Patient Success Rate	2/9	4/4	Meisel, Nasser, Ode, Stirling, Vieluf, Wang
Precision/PPV	2/9	1/4	Ode, Singh Rathore, Elemam, Wang
IoC / Time in Warning	0/9	2/4	Meisel, Nasser

5.4 Synthesis and Gaps

The concept matrix and detailed metrics reveal several patterns:

1. **Clinical translation gap:** Despite high sensitivities (>90% in several detection studies), FPR remains clinically unacceptable for standalone deployment. Only Dong et al. (0.165/h) and Fine (0.023/h) achieve FARs approaching clinical targets (<0.1–0.5/h) (Beniczky & Ryvlin, 2018).
2. **Personalization vs. generalization:** LOSO approaches achieve lower but more realistic performance (51.2% sensitivity, 43.5% patient success) compared to patient-specific tuning. Meisel et al.’s LOSO CV results suggest that aggregated performance metrics may overestimate real-world utility.
3. **Patient-level heterogeneity:** Patient success rates vary dramatically (43.5–100%), with Meisel and Ode showing that fewer than half of patients achieve usable performance. Nasser (5/6) and Stirling (100%) demonstrate that responder identification remains a critical challenge.
4. **Diary integration:** (Vieluf et al., 2025) demonstrates that hybrid approaches (wearable + clinical data) may outperform wearables alone, suggesting a direction for ambulatory systems where wearable-only performance is insufficient.
5. **Metric heterogeneity:** Inconsistent reporting (sensitivity vs. accuracy vs. AUC vs. IoC) impedes meta-analysis and clinical benchmarking. Key clinical metrics like detection latency and patient success rate are reported by fewer than half of studies.

6 Key Findings and Clinical Takeaways

This section synthesizes the evidence from 13 primary studies and 5 exploratory feasibility studies on wearable seizure detection and forecasting, identifying patterns that inform clinical practice and future research directions.

6.1 Executive Summary

- **Current State:** Wearable seizure detection achieves high sensitivity (71–100%) for convulsive seizures using accelerometry, but false alarm rates remain clinically unacceptable in most studies.
- **Forecasting Challenge:** Seizure forecasting remains immature, with only 43.5–83% of patients achieving better-than-chance performance and significant heterogeneity in individual responses.
- **Best Performing Approaches:** Multimodal sensor fusion (ACC + cardiac) and ensemble deep learning methods show the most promise for clinical translation.
- **Validation Gap:** Only 3 of 13 studies include prospective validation, with most relying on retrospective analyses that may overestimate real-world performance.

6.2 Detection Studies: Performance Landscape

6.2.1 High-Sensitivity Detectors

Table 4 summarizes the best-performing detection systems across different seizure types and settings.

Table 4: Highest Performing Detection Studies by Metric

Study	Seizure Type	Sensitivity	FAR	Latency	Validation	Ready
Fine 2025	Tonic	100%	0.023/h	14 s	Phase 1	○
Dong 2026	Nocturnal severe	71.6%	0.165/h	<30 s	Prospective	●
Spahr 2025	Convulsive	96%	<1/8d	26 s	Phase 2	●
Reintjes 2025	All types	98.16%	1.91/h	–	Retro	○
Borujeny 2013	Motor	100%	0	–	3-pt	○

Key Finding: Only two studies (Dong, Fine) achieve false alarm rates approaching clinical targets (<0.5/h) while maintaining reasonable sensitivity. The Phase 2 Spahr study demonstrates that multi-center validation is feasible but still shows FAR above clinical targets.

6.2.2 Clinical Translation Barriers

1. **False Alarm Crisis:** Reintjes et al. report FAR of 1.91–39.75/h, 20–200× higher than clinically acceptable. This confirms that ECG-only approaches without motion artifact filtering are not viable for standalone deployment.
2. **Seizure Type Limitation:** High sensitivity (>90%) is consistently achieved only for convulsive/motor seizures. Non-motor seizures remain largely undetectable by current wearable approaches.
3. **Offline Processing:** Most algorithms (Fine, Wang, Singh Rathore) run on offline servers rather than embedded devices, limiting real-time applicability.

6.3 Forecasting Studies: The Prediction Challenge

6.3.1 Performance Spectrum

Forecasting performance varies dramatically across studies and patients (Table 5).

Table 5: Forecasting Study Performance Comparison

Study	Metric	Value	TiW	Patient Success	Lead Time
Meisel 2020	IoC	14.1%	43.7%	43.5% (30/69)	–
Nasseri 2021	AUC-ROC	0.75	0.9–7.2 h/d	83% (5/6)	33 s mean
Stirling 2021	AUC	0.74	–	100% (hourly)	Cyclic HR
Vieluf 2025	Sensitivity	82%	–	–	Daily
Ode 2023	Sensitivity	74%	–	44% (29/66)	–

Critical Finding: Only Meisel et al. used true LOSO cross-validation, revealing that patient-specific tuning inflates performance by 30–40 percentage points. Their 43.5% patient success rate suggests that fewer than half of patients may benefit from current forecasting approaches.

6.3.2 The Responder Problem

- Meisel (2020): 43.5% significant forecasters
- Ode (2023): 44% patient success (29/66 with 100% sensitivity)
- Nasseri (2021): 83% success (5/6 above chance)
- Stirling (2021): 100% hourly success but highly variable AUC

Interpretation: Patient heterogeneity is the fundamental challenge in seizure forecasting. Universal models (Vieluf) may report aggregated performance that masks significant individual variation.

6.4 Methodological Insights

6.4.1 Algorithm Evolution Trajectory

Table 6: Evolution of Algorithms in Seizure Detection/Forecasting (2013–2026)

Period	Algorithms	Representative Studies	Characteristics
2013–2015	KNN, ANN, SVM	Borujeny 2013, Fujiwara 2016	Handcrafted features
2016–2019	HRV-based MLP, LSTM	Yamakawa 2020, Billeci 2018	Patient-specific models
2020–2022	LSTM+1DConv, Ensembles	Meisel 2020, Stirling 2021	LOSOS cross-validation
2023–2026	CNN+Attention, Matrix Profile, Self-Att AE	Ode 2023, Reintjes 2025, Spahr 2025	Anomaly detection paradigms

6.5 Exploratory Studies: Feasibility Evidence

6.5.1 Cardiac Signal Viability

The five exploratory studies collectively demonstrate that cardiac signals (ECG, HRV, PPG) contain detectable preictal changes:

- **Pavei 2017:** 94.1% sensitivity using SVM on HRV dynamics (n=12)
- **Fujiwara 2016:** 91% sensitivity using multivariate statistical process control
- **Leal 2017:** Identified RRMean and BPMMean as most consistent ECG features (n=167)
- **Jain 2017:** Demonstrated EDA decomposition feasibility for seizure-related autonomic changes
- **Seth 2023:** Systematic review showing 56–100% sensitivity range across 24 studies

Synthesis: Cardiac signals are viable for seizure detection but require sophisticated signal processing to overcome motion artifacts and individual variability.

6.6 Clinical Recommendations

6.6.1 For Current Clinical Practice

Based on current evidence, the following recommendations can be made:

1. **Nocturnal Monitoring:** Dong et al.’s NightWatch system (71.6% sensitivity, 0.165/h FAR) is the most validated option for home nocturnal monitoring.
2. **Tonic Seizures:** Fine et al.’s algorithm (100% sensitivity, 0.023/h FAR) shows promise for tonic seizure detection but requires Phase 2 validation.
3. **EMU Settings:** Spahr et al.’s Phase 2 validated algorithm (96% sensitivity, <1/8d FAR) is appropriate for EMU seizure counting.
4. **Avoid:** Standalone ECG-based detection systems (FAR >1/h) without confirmation from accelerometry.

6.6.2 For Researchers and Developers

1. **Validation Standard:** LOSO cross-validation should be the minimum standard for reporting performance.
2. **Metric Reporting:** All studies should report: sensitivity, specificity/PPV, FAR per hour, detection latency, and patient success rate.
3. **Interpretability:** SHAP or similar explainable AI frameworks should be incorporated to enable clinical trust and regulatory approval.
4. **Responder Identification:** Pre-study characterization of likely responders based on seizure type, HRV reactivity, and other biomarkers.

6.7 Limitations of Current Evidence

6.7.1 Study Design Limitations

- **Small Samples:** Median $n=15-70$ patients; most studies underpowered for subgroup analysis
- **Short Duration:** Median monitoring <1 week per patient; rare events (seizures) limit statistical power
- **Retrospective Bias:** 10/13 studies retrospective; high risk of data leakage and optimistic bias
- **Setting Bias:** Most studies in EMU; home performance likely worse

6.7.2 Metric Heterogeneity

- Sensitivity vs. accuracy vs. AUC reported inconsistently
- FPR reported as per-hour, per-day, per-night, or not at all
- Patient success rate reported by only 6/13 studies
- No standardized benchmark datasets (except EPILEPSIAE)

6.8 Future Directions

6.8.1 Research Priorities

Based on identified gaps, the following research priorities emerge:

1. **Standardized Metrics:** Develop consensus on minimum reporting standards (ILAE Phase-appropriate metrics)
2. **Responder Prediction:** Identify baseline biomarkers that predict which patients will achieve usable forecasting performance
3. **Edge Optimization:** More studies like Spahr (1.12 s inference) needed for real-time deployment
4. **Personalization at Scale:** Federated learning approaches that enable patient-specific models without data centralization
5. **Hybrid Systems:** Combined wearable + diary approaches (Vieluf) that leverage complementary information sources

6.8.2 Emerging Technologies

- **Matrix Profile Anomaly Detection:** Reintjes et al. show unsupervised approaches can achieve high sensitivity without seizure-specific training
- **Self-Attentive Autoencoders:** Ode et al. demonstrate AE approaches for RRI-based forecasting

- **Attention Mechanisms:** Dong et al. use attention to weight reliable signals and reduce motion artifact impact
- **Ensemble Deep Learning:** Spahr et al. show ensemble CNN approaches achieve Phase 2 validation

6.9 Conclusion

Wearable seizure detection and forecasting have advanced significantly from proof-of-concept studies to Phase 2 clinical validation. Detection of convulsive seizures is now reliable enough for limited clinical use, with two systems (Dong, Fine) achieving false alarm rates compatible with patient adoption. However, seizure forecasting remains an open challenge, with fewer than half of patients achieving better-than-chance performance in LOSO-validated studies.

The path forward requires: (1) rigorous LOSO validation as the standard, (2) standardized metric reporting enabling cross-study comparison, (3) interpretable models that earn clinician trust, and (4) recognition that patient heterogeneity may require personalized approaches rather than universal solutions. The integration of multimodal sensors, attention mechanisms, and explainable AI represents the current state-of-the-art, but clinical translation demands prospective, multi-center validation in real-world settings.

7 Conclusion

This systematic review examined thirteen studies on deep learning approaches for ambulatory seizure monitoring using non-EEG biosignals, encompassing both detection and forecasting applications published between 2013 and 2026. The evidence reveals a field at a crossroads: detection technology has achieved substantial technical maturity for convulsive seizures, yet critical barriers prevent widespread clinical adoption. Below, we synthesize the current state of methodological development, identify persistent translation challenges, and outline priorities for future research.

7.1 Current State of the Field

The field of ambulatory seizure monitoring using non-EEG biosignals has achieved substantial technical maturity in detection capabilities while forecasting remains an emerging challenge. Across nine detection studies, deep learning systems demonstrate high sensitivity ranging from 71.6% to 100% for identifying convulsive seizures, with accelerometry-based approaches proving particularly effective for motor manifestations. This performance foundation establishes that wearable devices can reliably detect generalized tonic-clonic and other convulsive seizure types in controlled settings. However, false alarm rates reveal a critical translation gap: only two studies achieve clinically acceptable performance below 0.5 false alarms per hour. Fine et al. report 0.023/h for tonic seizures, while Dong et al. achieve 0.165/h for nocturnal severe seizures, representing the only systems approaching deployable thresholds.

Seizure forecasting exhibits markedly less maturity than detection. Across four prospective studies, only 43.5% to 83% of patients demonstrate better-than-chance performance, highlighting a fundamental responder problem that limits universal applicability. This

patient heterogeneity means that aggregated performance metrics obscure substantial individual variation. When rigorous leave-one-subject-out validation is applied, as in Meisel et al., fewer than half of patients achieve significant forecasting capability. The forecasting challenge thus appears less tractable than detection in the current state of development.

Sensor modality effectiveness follows a clear hierarchy. Accelerometry dominates the evidence base, appearing in eight of thirteen studies and providing the most robust signal for convulsive seizure detection. Cardiac signals contribute value in six studies, particularly for preictal changes and forecasting applications, but standalone electrocardiogram approaches suffer from excessive motion artifact sensitivity and false alarm rates. Multimodal fusion strategies, combining accelerometry with cardiac or electrodermal measures in four recent studies, represent the current frontier for improving specificity while maintaining sensitivity.

The clinical utility of current systems remains tightly bounded by seizure semiology. Convulsive seizures with prominent motor components are highly detectable through wrist-worn accelerometry, achieving sensitivity above 90% in multiple studies. By contrast, non-motor seizures—including focal aware seizures and absence seizures—remain largely undetectable through available wearable modalities. This detection gap creates a fundamental limitation: patients whose seizure semiology lacks convulsive features cannot benefit from current ambulatory monitoring approaches, regardless of algorithmic sophistication.

Validation rigor represents another limiting factor in the field’s current state. Only four of thirteen studies employ leave-one-subject-out or patient-independent cross-validation, with most relying on retrospective analyses that likely overestimate real-world performance. Prospective validation remains rare, and multi-center evidence is limited. The evidence base thus reflects promising but early-stage technology, where algorithmic innovation has outpaced rigorous clinical validation in diverse ambulatory settings.

7.2 Methodological Evolution

The methodological landscape of wearable seizure detection and forecasting has undergone substantial transformation from 2013 to 2026, reflecting both technical advances in deep learning and increasing rigor in clinical validation. This evolution occurred in three interconnected dimensions: algorithmic sophistication, validation methodology, and clinical readiness assessment.

Early work from 2013 to 2015 relied predominantly on traditional machine learning approaches. Studies during this period employed k-nearest neighbors, artificial neural networks, and support vector machines operating on handcrafted features extracted from cardiac signals. These methods, while interpretable, required extensive domain expertise for feature engineering and typically achieved moderate performance on small datasets.

The period from 2016 to 2019 marked a transition toward heart rate variability-based deep learning models, particularly multilayer perceptrons and long short-term memory networks. This shift enabled automated feature extraction from temporal HRV patterns, though these studies often relied on patient-specific hyperparameter tuning that inflated apparent performance. The architectural advances during this era prioritized sensitivity over specificity, resulting in unacceptably high false alarm rates.

From 2020 to 2022, algorithm development emphasized architectural complexity through hybrid models combining LSTM networks with one-dimensional convolutional layers, alongside ensemble methods. These architectures explicitly addressed temporal dependen-

cies in cardiac and accelerometric signals while improving robustness through ensemble diversity. The introduction of leave-one-subject-out cross-validation during this period revealed that patient-specific tuning had previously obscured substantial performance variation across individuals.

The most recent period from 2023 to 2026 has witnessed the emergence of attention-based architectures, matrix profile anomaly detection, and self-attentive autoencoders. Attention mechanisms enable models to selectively weight informative signal segments while suppressing motion artifacts and noise. Unsupervised approaches have demonstrated that seizure-related autonomic changes can be detected without requiring large annotated seizure datasets, potentially addressing the chronic data scarcity problem.

Validation methodology has evolved more slowly than algorithm architecture, creating a persistent gap between technical performance and clinical applicability. Early studies relied on simple train-test splits without subject-level separation, enabling data leakage that severely overestimated real-world performance. The recognition of this problem led to widespread adoption of LOSO cross-validation, which provides a more realistic assessment of generalizability to new patients. Despite this improvement, prospective validation remains rare, with most studies relying on retrospective analyses.

The International League Against Epilepsy Phase framework provides a structured approach to assessing clinical readiness. The current evidence base remains concentrated in Phases 1 and 2, with only two studies achieving Phase 2 validation. No studies have progressed to Phase 3 or beyond, indicating that wearable seizure detection and forecasting remain in early translational stages despite more than a decade of research.

A persistent obstacle to progress is the extreme heterogeneity in performance metrics reporting. Studies report sensitivity, specificity, accuracy, precision, recall, F1-score, and area under the ROC curve with no consensus on primary endpoints. False alarm rates, the most clinically relevant metric for patient adherence, are reported in incompatible units or omitted entirely. This diversity impedes direct comparison across studies and meta-analytic synthesis. The absence of standardized benchmark datasets further compounds this problem.

The relationship between algorithmic sophistication and clinical readiness follows a counterintuitive trajectory. Architectural advances have consistently outpaced validation rigor, with studies employing state-of-the-art deep learning models yet failing to demonstrate prospective validity. This disconnect suggests that research effort has been misallocated toward incremental architectural improvements rather than addressing fundamental validation challenges.

7.3 Barriers to Clinical Translation

Four interrelated barriers prevent widespread clinical adoption of deep learning systems for ambulatory seizure monitoring: false alarm rates, methodological limitations in validation studies, patient heterogeneity, and scope restrictions regarding detectable seizure types.

The most immediate technical barrier is the unacceptable false alarm rate reported in most studies. While detection sensitivity consistently exceeds 90% for convulsive seizures, only two of thirteen detection studies achieve false alarm rates within clinically tolerable targets. The remaining systems generate 20 to 200 times more false alarms than clinicians and patients can accept, with ECG-based approaches reaching 1.91 to 39.75 false alarms per hour. This discrepancy means that most validated algorithms would generate 50 to 500 false alarms per day in home settings, ensuring patient non-adoption and caregiver

fatigue. The single exception among cardiac-based systems is Fine’s tonic seizure detector, which demonstrates that aggressive artifact rejection and narrow seizure focus can achieve clinical viability, albeit at the cost of limited generalizability.

Three systematic methodological weaknesses inflate reported performance and limit generalizability. First, sample sizes are critically underpowered, with median cohort sizes of 15 to 70 patients. These small samples preclude meaningful subgroup analysis by seizure type, medication regime, or comorbidities. Second, monitoring duration is typically limited to days or weeks rather than months, which biases performance estimates toward common seizure types while underestimating challenges with rare events. Third, setting bias is pervasive: eleven of thirteen studies conduct validation in epilepsy monitoring units under controlled conditions that minimize motion artifacts, environmental noise, and device compliance problems. The two studies that tested systems in actual home environments both reported performance degradation compared to EMU results, suggesting that current validation standards substantially overestimate real-world utility.

The most fundamental barrier to clinical translation is the responder problem: current forecasting and detection approaches work well for a minority of patients but fail for the majority. In leave-one-subject-out cross-validation, which represents true generalization to unseen patients, only 43.5% of patients achieved better-than-chance forecasting performance. Similar patterns appear in detection studies, where patient success rates range from 44% to 100% across studies. This heterogeneity means that aggregated performance metrics obscure the reality that clinicians cannot predict in advance which patients will benefit from wearable monitoring. The absence of baseline biomarkers to identify likely responders forces a trial-and-error approach that is incompatible with clinical practice.

A final barrier is the restricted seizure type coverage of current non-EEG approaches. Accelerometry-based systems reliably detect convulsive and motor seizures but cannot identify focal aware seizures, absence seizures, or other non-motor events that constitute the majority of seizures in many patients. Cardiac-based approaches show promise for preictal detection but are limited by motion artifact sensitivity and the fact that only 60 to 70% of patients exhibit reliable heart rate changes before seizures. This scope limitation means that even a perfectly optimized wearable system would remain incomplete without complementary EEG or multimodal integration, restricting clinical utility to a subset of the epilepsy population.

7.4 Future Directions

The transition from research prototypes to clinically viable seizure monitoring systems requires addressing several methodological and implementation gaps. Based on our findings, we identify three priority areas that warrant immediate attention.

Standardization and benchmarking represent the foundation for progress. The field lacks consensus on evaluation metrics and reporting standards, making cross-study comparisons difficult. Minimum reporting requirements should include patient-level confusion matrices, exact seizure durations, latency distributions, and battery consumption metrics. Open benchmark datasets with annotated non-motor seizures would enable meaningful algorithm comparisons. Additionally, standardized protocols for simulating real-world conditions—sensor displacement, signal artifacts, and data loss—would better prepare systems for deployment.

Real-world deployment challenges must be addressed prospectively. Most validated algorithms remain unevaluated in home environments. Prospective studies must assess

long-term reliability across diverse seizure types, daily activities, and patient populations. Edge optimization represents a critical bottleneck: while deep learning models demonstrate strong performance, computational demands often exceed embedded device constraints. Research should explore model compression techniques, lightweight architectures, and energy-efficient inference strategies. Integration with wearable hardware requires addressing signal synchronization, multi-modal data fusion, and continuous operation over extended periods.

Personalization and interpretability are essential for clinical acceptance. Universal models rarely achieve optimal performance across all patients due to high inter-individual variability. Federated learning approaches offer a pathway to personalized models without centralizing sensitive data, preserving privacy while enabling adaptation. Simultaneously, explainable AI methods must bridge the gap between high accuracy and clinical trust. Visual explanations of feature contributions, uncertainty quantification, and interpretable error analysis would facilitate clinician acceptance and regulatory approval.

Expanding detection scope beyond convulsive seizures remains a critical unmet need. Current research predominantly addresses motor seizures, leaving absence seizures and focal impaired awareness seizures largely unaddressed. Future work should develop multi-modal approaches that combine physiological signals with behavioral markers. Hybrid systems integrating wearable data with patient-specific information—medication schedules, seizure diaries, and comorbidities—may improve sensitivity while reducing false alarms.

The path forward requires collaboration between machine learning researchers, clinicians, and industry partners. Priority should be given to prospective validation studies that report both technical and patient-centered outcomes, ensuring that advances translate into meaningful improvements in epilepsy management.

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